

Daniel Heraldito Certuche Grueso

Data Scientist. Applied machine learning and generative AI

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Professional profile

My work as a data scientist combines a background in astronomy with industrial experience building solutions on real-world data. Research taught me to work with weak signals, measure uncertainty, and avoid turning a correlation into an easy conclusion. In industry, I have applied that discipline to segmentation and anomaly detection models, semantic recommendation, and generative AI for document processing. I also contribute to preparing and delivering the data behind these systems, because an analytical solution creates value only when its data is reliable, its results are reproducible, and people can use it.

Professional experience

Data Engineer and Data Scientist

Crystal S.A.S., Medellín

Jul 2025–present

- Ship LLM features into applications the business uses: natural-language search that turns a question into structured filters, validated against a closed vocabulary before they reach the query. The prompt travels as a bound parameter, calls run under a timeout, and a deterministic parser takes over when the model fails.
- Built a tool-calling assistant directly against the model provider API, with no orchestration framework: a bounded reasoning loop, reads in parallel and writes one at a time, and a confirmation step that states what will change before any write. Agents reach company data through read-only MCP servers I wrote, with object allowlists and row limits.
- Built semantic retrieval on embeddings from a serving endpoint indexed in MongoDB Atlas, reusing the same text template when indexing and when querying so ranking stays consistent, and validated the live index before release.
- Develop segmentation models with K-Means and principal component analysis, and anomaly detection with Isolation Forest and LOF, registered in MLflow on Unity Catalog with controlled promotion across environments.
- Prepare data on a governed four-layer lakehouse with PySpark and Spark SQL transformations, idempotent loads, and control tables so retraining runs start from a reproducible data foundation.

Data Science Intern

Crystal S.A.S., Medellín

Jan–Jun 2025

- Automated cloud data pipelines in Python, PySpark, and SQL across scattered operational sources, and built a semantic product recommendation flow integrated with SQL Server.

Industry project

Batch document understanding with LLMs

Same employer, collaborative project

Apr 2026–present

- Structured the portable, version-controlled form of a Databricks pipeline that ingests PDF files from SFTP into Unity Catalog, identifies files by content, and parses them with native AI functions.
- Wrote the per-document-type prompts and their output contracts, and contributed to extraction into Delta tables with validation and audit queries.

Technical skills

LLM applications and agents. Native AI functions on Databricks for document parsing and inference over serving endpoints, Meta Llama 3.3 70B and hosted embedding models, Azure OpenAI over its REST interface, tool calling with JSON schemas, prompt design and versioning by document type, output validation and graceful fallback, MCP servers in Python and TypeScript, MLflow on Unity Catalog. Retrieval on MongoDB Atlas with a vector index.

Statistical and modeling methods. Regression and correlation, significance testing, outlier treatment, dimensionality reduction, clustering and validation, anomaly detection, and method robustness analysis.

Languages and platform. Python and SQL. PySpark, Spark SQL, Pandas, NumPy and SciPy. Databricks, Delta Lake, Unity Catalog, SQL Warehouse, Jobs and Asset Bundles. SQL Server, Azure SQL and MongoDB Atlas. FastAPI, React, Streamlit, Databricks Apps, Docker, Git and Azure DevOps.

Education and languages

B.Sc. in Astronomy

University of Antioquia, Medellín

Oct 2025

Coursework in Astrostatistics and in Data Analytics with AI.

B.Sc. thesis. *Morphological Characterization of Spiral Arms in Disk Galaxies from the IllustrisTNG50 Simulation and Their Relation to the Properties of Their Host Dark Matter Halos.*

Languages. Spanish: native; English: B2 CEFR, used professionally in technical contexts.

Additional experience

Science Educator and Facilitator

Cosmo Education S.A.S., Medellín

Feb 2024–Jan 2025

- Designed and facilitated hands-on science and astronomy learning experiences.

Science Communicator

Parque Explora, Medellín

Aug 2019–Jun 2022

- Led museum and planetarium workshops, translating complex science into clear experiences for varied audiences.